

Satellite Realignment System

Operational Report (with IMU Feedback)

Generated: 2026-05-05 11:08 UTC

1. System Overview

Parameter	Value
Stations Monitored	2
Online Stations	2
IMU Feedback	Active (MPU-6050)
Architecture	ESP32 → FastAPI → Dashboard
Report Timestamp	2026-05-05 11:08 UTC

2. Station Status

Station	Status	Mode	AZ (°)	EL (°)	IMU AZ (°)	IMU EL (°)	Signal (dBm)	IMU Cal
Station A – Harare	ONLINE	Mode.AUTO	65.0	111.5	0.0	0.0	-6.0	No
Station B – Victoria Falls	ONLINE	Mode.AUTO	104.0	77.5	0.0	0.0	-9.0	No

3. Movement Verification Log

Time	Station	Result	IMU AZ Δ (°)	IMU EL Δ (°)	Target AZ	Target EL
2026-05-04T22:56	Station A – Harare	FAIL	7.91	0.00	0.0°	127.9°
2026-05-04T22:55	Station A – Harare	FAIL	35.94	0.00	0.0°	127.9°

4. Position vs Signal Log

Time	Station	AZ (°)	EL (°)	Signal (dBm)
2026-05-05T11:08	Station B – Victoria Falls	104.0	77.5	-10.0
2026-05-05T11:08	Station A – Harare	65.0	111.5	-6.0
2026-05-05T11:07	Station B – Victoria Falls	104.0	77.5	-9.0
2026-05-05T11:07	Station A – Harare	65.0	111.5	-5.0
2026-05-05T11:07	Station A – Harare	65.0	111.5	-6.0
2026-05-05T11:07	Station B – Victoria Falls	104.0	77.5	-9.0
2026-05-05T11:07	Station A – Harare	65.0	111.5	-6.0
2026-05-05T11:07	Station A – Harare	65.0	111.5	-5.0
2026-05-05T11:07	Station B – Victoria Falls	104.0	77.5	-9.0
2026-05-05T11:07	Station B – Victoria Falls	104.0	77.5	-9.0
2026-05-05T11:07	Station A – Harare	65.0	111.5	-6.0

2026-05-05T11:07	Station B – Victoria Falls	104.0	77.5	-9.0
2026-05-05T11:07	Station A – Harare	65.0	111.5	-5.0
2026-05-05T11:07	Station A – Harare	65.0	111.5	-5.0
2026-05-05T11:07	Station B – Victoria Falls	104.0	77.5	N/A
2026-05-05T11:07	Station B – Victoria Falls	104.0	77.5	N/A
2026-05-05T11:07	Station A – Harare	65.0	111.5	-5.0
2026-05-05T11:07	Station A – Harare	65.0	111.5	-6.0
2026-05-05T11:07	Station B – Victoria Falls	104.0	77.5	-10.0
2026-05-05T11:07	Station B – Victoria Falls	104.0	77.5	-10.0
2026-05-05T11:07	Station A – Harare	65.0	111.5	-6.0
2026-05-05T11:07	Station A – Harare	65.0	111.5	-6.0
2026-05-05T11:07	Station B – Victoria Falls	104.0	77.5	-10.0
2026-05-05T11:06	Station B – Victoria Falls	104.0	77.5	-10.0
2026-05-05T11:06	Station A – Harare	65.0	111.5	-5.0
2026-05-05T11:06	Station A – Harare	65.0	111.5	-6.0
2026-05-05T11:06	Station B – Victoria Falls	104.0	77.5	-10.0
2026-05-05T11:06	Station B – Victoria Falls	104.0	77.5	-10.0
2026-05-05T11:06	Station A – Harare	65.0	111.5	-5.0
2026-05-05T11:06	Station B – Victoria Falls	104.0	77.5	-10.0
2026-05-05T11:06	Station A – Harare	65.0	111.5	-5.0
2026-05-05T11:06	Station A – Harare	65.0	111.5	-5.0
2026-05-05T11:06	Station B – Victoria Falls	104.0	77.5	-10.0
2026-05-05T11:06	Station A – Harare	65.0	111.5	-6.0
2026-05-05T11:06	Station B – Victoria Falls	104.0	77.5	-10.0
2026-05-05T11:06	Station B – Victoria Falls	104.0	77.5	-9.0
2026-05-05T11:06	Station A – Harare	65.0	111.5	-6.0
2026-05-05T10:05	Station A – Harare	150.0	120.2	-11.0
2026-05-05T10:05	Station B – Victoria Falls	65.0	160.0	-14.0
2026-05-05T09:59	Station A – Harare	150.0	120.2	N/A
2026-05-05T09:59	Station A – Harare	65.0	160.0	N/A
2026-05-05T09:59	Station A – Harare	150.0	120.2	N/A
2026-05-05T09:59	Station A – Harare	65.0	160.0	N/A
2026-05-05T09:59	Station A – Harare	150.0	120.2	N/A
2026-05-05T09:59	Station A – Harare	65.0	160.0	N/A
2026-05-05T09:59	Station A – Harare	150.0	120.2	N/A
2026-05-05T09:59	Station A – Harare	65.0	160.0	N/A
2026-05-05T09:59	Station A – Harare	150.0	120.2	N/A
2026-05-05T09:58	Station A – Harare	65.0	160.0	N/A
2026-05-05T09:58	Station A – Harare	150.0	120.2	N/A

4. Environmental Data

Station	Timestamp	Temp (°C)	Wind (km/h)	Rain	Humidity	Pressure
Station_2	2026-05-05T11:07:50.679496+00:00	30.6	12.1	0.0	38%	916
	2026-05-05T11:07:49.349794+00:00	25.1	11.5	0.0	37%	859
Station_1	2026-05-05T11:07:09.413663+00:00	30.6	12.1	0.0	38%	916
	2026-05-05T11:07:08.097629+00:00	25.1	11.5	0.0	37%	859
Station_2	2026-05-05T11:06:31.439833+00:00	25.1	11.5	0.0	37%	859
	2026-05-05T11:05:55.308497+00:00	30.6	12.1	0.0	38%	916
Station_1	2026-05-05T11:05:54.048545+00:00	25.1	11.5	0.0	37%	859
	2026-05-05T11:04:16.269457+00:00	30.6	12.1	0.0	38%	916
Station_2	2026-05-05T11:04:16.257569+00:00	25.1	11.5	0.0	37%	859
	2026-05-05T11:00:52.632619+00:00	30.6	12.1	0.0	38%	916
Station_1	2026-05-05T11:00:50.022453+00:00	25.1	11.5	0.0	37%	859
	2026-05-05T10:59:16.278106+00:00	30.5	11.8	0.0	39%	916
Station_2	2026-05-05T10:59:16.259780+00:00	25.0	11.3	0.0	37%	859
	2026-05-05T10:55:47.120936+00:00	30.5	11.8	0.0	39%	916
Station_1	2026-05-05T10:55:45.797532+00:00	25.0	11.3	0.0	37%	859
	2026-05-05T10:54:16.407574+00:00	30.5	11.8	0.0	39%	916
Station_2	2026-05-05T10:54:16.393949+00:00	25.0	11.3	0.0	37%	859
	2026-05-05T10:50:44.420074+00:00	30.5	11.8	0.0	39%	916
Station_1	2026-05-05T10:50:43.109087+00:00	25.0	11.3	0.0	37%	859
	2026-05-05T10:49:16.288043+00:00	30.5	11.8	0.0	39%	916

5. System Logs (Last 30)

Timestamp	Station	Level	Message
2026-05-05T11:08	system	INFO	Backend v3 started — IMU feedback active
2026-05-05T11:08	station_2	INFO	Station initialised: station_2
2026-05-05T11:08	station_1	INFO	Station initialised: station_1
2026-05-05T11:08	system	INFO	Backend v3 started — IMU feedback active
2026-05-05T11:08	station_2	INFO	Station initialised: station_2
2026-05-05T11:08	station_1	INFO	Station initialised: station_1
2026-05-05T11:07	system	INFO	Backend v3 started — IMU feedback active
2026-05-05T11:07	station_2	INFO	Station initialised: station_2
2026-05-05T11:07	station_1	INFO	Station initialised: station_1
2026-05-05T11:07	system	INFO	Backend v3 started — IMU feedback active
2026-05-05T11:07	station_2	INFO	Station initialised: station_2
2026-05-05T11:07	station_1	INFO	Station initialised: station_1
2026-05-05T11:07	system	INFO	Backend v3 started — IMU feedback active

2026-05-05T11:07	station_2	INFO	Station initialised: station_2
2026-05-05T11:07	station_1	INFO	Station initialised: station_1
2026-05-05T11:07	system	INFO	Backend v3 started — IMU feedback active
2026-05-05T11:07	station_2	INFO	Station initialised: station_2
2026-05-05T11:07	station_1	INFO	Station initialised: station_1
2026-05-05T11:07	system	INFO	Backend v3 started — IMU feedback active
2026-05-05T11:07	station_2	INFO	Station initialised: station_2
2026-05-05T11:07	station_1	INFO	Station initialised: station_1
2026-05-05T11:07	system	INFO	Backend v3 started — IMU feedback active
2026-05-05T11:07	station_2	INFO	Station initialised: station_2
2026-05-05T11:07	station_1	INFO	Station initialised: station_1
2026-05-05T11:07	system	INFO	Backend v3 started — IMU feedback active
2026-05-05T11:07	station_2	INFO	Station initialised: station_2
2026-05-05T11:07	station_1	INFO	Station initialised: station_1
2026-05-05T11:07	system	INFO	Backend v3 started — IMU feedback active
2026-05-05T11:07	station_2	INFO	Station initialised: station_2
2026-05-05T11:07	station_1	INFO	Station initialised: station_1